than one option on the table then how do you decide?" asks Doreen Fraser, a philosopher of physics at the University of Waterloo. "Is it because you have certain prejudices about what's interesting and what's not interesting? Actually that's a large part of it, but that's kind of uncomfortable." This discomfort, this fear, is yet another reason why it's tempting, as a physicist, to "shut up and calculate." But giving in to that fear just makes it harder for us to see our biases.

These biases include a lot of the factors we've seen directly at play in this book: political considerations, funding models, the milieu of ideas in particular places and times, even simple interpersonal disputes. There are also many biases that have been at work throughout this book, but not in the foreground. A woman, Grete Hermann, found the problem with von Neumann's proof thirty years before Bell did, but nobody noticed at the time. It's hard to imagine that her gender had nothing to do with the reception of her work in 1935, a time when women were still generally not allowed to teach at universities. And, given that specialization in the foundations of quantum physics was a professional strike against any physicist, it's also not far-fetched to imagine that few women and nonwhite physicists would be attracted to work in that area, since their identities were already a strike against them within science and academia as a whole, because of systemic bias. This would explain the otherwise remarkable fact that hardly any women or people who aren't white appear anywhere in this story at all. Prejudices, against certain kinds of people as well as certain kinds of ideas, are pervasive throughout science.

But the mere existence of these biases doesn't mean that science is identical to all those other spheres of human endeavor or that scientific truth isn't different from ill-informed opinions with no connection to experiment or reality. Our biases don't fully determine the content of our best scientific theories—reality pushes back, and we want it to push back as hard as we can allow it. That pushback constrains the possible hypotheses we entertain as scientists. There's a wide middle ground between "science is Pure and Perfectly Rational" and "science is just some bullshit somebody made up." There's still plenty of room for humans to interfere in that middle ground, as we've seen throughout this book. But that doesn't mean science isn't to be trusted—that's as naive as the Sherlock Holmes view of science.

That being said, the story of quantum foundations does seem to call into